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Towards the quantitative control of crop production and quality. III. Some recent developments in research into the root-soil interface

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TOWARDS THE QUANTITATIVE CONTROL OF CROP PRODUCTION AND QUALITY. III. SOME RECENT DEVELOPMENTS IN RESEARCH INTO THE ROOT-SOIL INTERFACE

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ABSTRACT: In addition to their general role as sinks for ion transfer from the soil, roots can modify uptake by specifically changing conditions within the rhizosphere. The possibility of including such factors in models is illustrated by the effects of pH change, solubilization of iron, mycorrhizal infections, competitive ion uptake and root shrinkage caused by drying.

INTRODUCTION

In preparing this lecture, I found that I was rather overwhelmed by the amount of interesting material available, since the micro-environment of the root-soil interface challenges the skills of a fair proportion of the specialists needed to explain the workings of the macro-environment itself. I finally decided to mention five topics that struck me as having important novel features which have not yet received as much attention as they deserve.

In the 1960s and 1970s, it was shown how plant nutrients are transported to root surfaces by diffusion and convection; and how their concentration profiles in the rhizosphere could be both measured and predicted. In the picture that emerged the root played the role of a cylindrical sink, which absorbed nutrients, but did not otherwise influence its immediate environment. In contrast, in all the topics I shall mention, the root has an additional and very active role, though in each instance it is a different one.

pH Chances in the Rhizosphere

From the standpoint of nutrition the most important way in which the root

TABLE 3.1. Some Effects Of Rhisophere pH.

Effect	Example
Conc. major nutrient	$H_2PO_4^-$
Conc. micro-nutrient	Fe^{3+}
Conc. toxic acid	$Al^{3+}(6H_2O)$
Conc. toxic base	NH_3 aq
Rate of N_2 fixation	Rhizobium
Bacterial growth	Rhizobium
Root infection by bacteria	Rhizobium
Nitrification	Nitrosomonas
Bacterial-fungi balance	
Distribution of fungi	
Lysis of hyphae	
Enzyme activity	

changes its environment is by altering its pH. The effects are legion. Table 3.1 lists some of the most significant. Immediately, this raises the possibility that we may be able to manipulate plants by breeding or fertilizer management to provide a favourable pH regime; but to do this we have to understand the sources of acids or bases released from the root surfaces (1). These are as follows: (i) imbalance between nutrient cation and anion uptake across the root-soil interface; (ii) CO_2 respiration; (iii) excretion of organic acids from the root; and (iv) microbial production of acids from root carbon release. By far the most important sources are the hydrogen or bicarbonate ions. These are produced to maintain electrical neutrality across the roots' interface with the soil when it is absorbing more nutrient cations than anions or anions than cations. This balance is dominated by the form in which nitrogen enters the root. If it is as nitrate, bicarbonate is exported and the pH will rise. If it is as ammonium, the hydrogen ion is exported and the pH will fall. If it is as urea or uncharged molecular nitrogen, as in nitrogen fixation, there is also an appreciable, but smaller export of hydrogen ion (Fig. 3.1). These are major effects, amounting, for nitrate intake, to as much as one

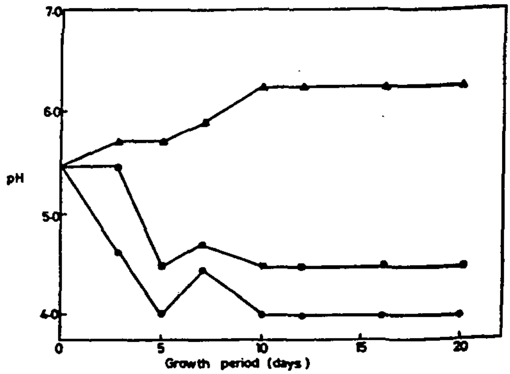

Figure 3.1 pH changes in the nutrient media during the growth of tomato plants (pH adjusted back to 5.5 after every determination (▲) NO_3, (■) Urea, (●) NH_4 (after Kirkby, 1969).

third of the total salt intake, and even more in the case of ammonium intake. For example Cunningham (3) found that, on an average over 62 common arable crop and weed species, the uptake of anions (including nitrate) and cations was 3.6 and 2.5 mmol charge per g dry weight, with a consequent export of 1.1 mmol HCO_3^- per gram dry weight. On the other hand, for N-fixing leguminous crops, Nyatsanga and Pierre (4) found alfalfa exported 0.8 mmol H^+ per g dry weight into soil, and Israel and Jackson (5) found nodulated soya to release 1.08 mmol H^+ per g dry weight into nutrient solution. In most arable conditions roots take in nitrogen mainly as nitrate. The consequent export of bicarbonate largely depends on the plant species. Plants having their nitrate reductase system mainly in their roots tend to export the basic hydroxyl ion produced in the reduction directly, after reaction with carbon dioxide, as bicarbonate. Monocots, especially grasses, are in this category. If the nitrogen reductase system is in the tops much of the hydroxyl ion produced is converted into carboxylic anions, such as malate or oxalate, which are accumulated in cell vacuoles (6). Consequently these species have much less tendency to raise the soil pH, and can indeed be influenced to lower it. Buckwheat, *Fagopyrum esculentum*, is a notable example.

The other potential sources of acids and bases are quantitatively far less important (1). The rise in pressure of carbon dioxide close to the root surface has been calculated to be very small indeed in aerobic soils through which the gas can diffuse freely. Roots certainly exude organic substances, but only small amounts of organic acids can be among them since the cytoplasm of root cells is fairly closely controlled at pH between 6 and 7, which is well above the pK of any potential organic acids. These will therefore be exported as their anions. By making the very generous assumption that one third of the total carbon (excluding carbon dioxide) exuded by roots is processed by acid forming bacteria in the rhizosphere to form acetic acid or other low molecular weight acids it may be calculated that 1 mmol charge/g plant dry weight could be produced - a figure comparable with the acid released by plants fixing their own nitrogen. However, nothing approaching these amounts has ever been recovered in the soil, despite diligent search (Table 3.2).

The effect of acid or base release into the soil may be to shift the pH at the root surface by as much as two units. This has been measured experimentally (8), and calculated using the theory of the diffusion of acids and bases through the soil (1). This theory enables one to calculate the pH profile across the rhizosphere in terms of such variables as the pH buffer capacity of the soil, its moisture level, and initial pH, and the rate of release of acid at the root surface (Figures 3.2 and 3.3).

I have mentioned that the effects of pH change in the rhizosphere are legion, and I referred in my last lecture to the importance of nitrogen fixing legumes in solubilizing rock phosphate. Figure 3.4 shows this effect. Here I will give another less obvious example. Some ten years ago we set about finding out why rape, *Brassica napus*, would thrive in phosphate deficient soils in which other plants, like onion, would die. We found that phosphate deficient rape was actually acidifying its rhizosphere in spite of the fact that it was taking up its nitrogen as nitrate. The acid released solubilized soil phosphorus. Rape supplied with phosphate did not show this effect (Fig. 3.5). Here appeared to be an adaptive plant survival mechanism, worth investigating further. In solution culture we found that rape roots deprived of nitrate instantly ceased exporting bicarbonate (Fig. 3.6). If deprived of phosphate there was little effect for about three days and then they tended to release acid. Study of the cation-anion balances showed that when deprived of phosphate, uptake of nitrate is more severely affected than uptake of cations (Table 3.3) Consequently, the usual export of bicarbonate is

TABLE 3.2. Amounts of Organic Acids and Anions Extracted from Rhizosphere and Non-rhizosphere Begbroke Sandy Loam [after (6)].

Method	Extractable anion concentration (μmol charge/g soil) Non-rhizosphere soil	Rhizosphere soil
Resin extraction (for non-volatile acids excluding oxalic)	12.0±1.1	9.3±0.5
Steam distillation (for volatile acids + CO_2)	9.3±0.5	8.5±0.5
Alkaline-ethanol extraction (for oxalate)	9.3±1.0	12.0±0.5

TABLE 3.3. Effects of Withdrawing P on Cation/Anion Uptake by Rape [after (11)].

	Uptake (meq/g d.wt.) day 15-32				
	K^+	Na^+	Ca^{2+}	Mg^{2+}	ΣC^+
+P	0.59	0.002	1.18	0.27	2.04
-P	0.20	0.001	1.04	0.24	1.46
	NO_3^-	$H_2PO_4^-$	SO_4^{2-}	Cl^-	ΣA^-
+P	2.27	0.15	0.42	0.90	3.73
-P	1.07	0	0.29	0.30	1.65
	$\Sigma A^- - \Sigma C^+$				
+P	1.68				
-P	0.19				

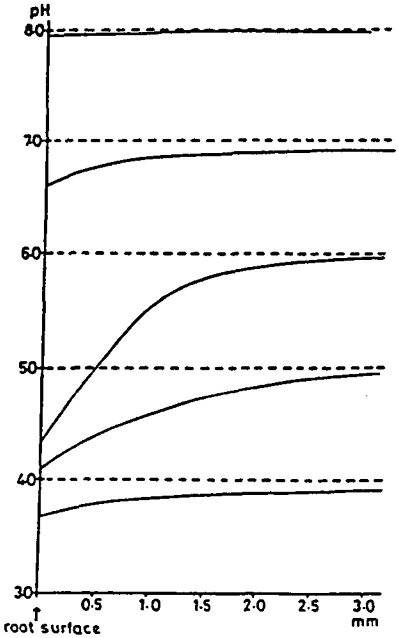

Figure 3.2 Effect of initial soil pH on pH profiles following H^+ release at the root surface (time, 10^6s) (after Nye, 1981). Diffusion of acids and bases through the soil is slow between pH 5.0 and 6.0 so that pH gradients are steeper in this range.

greatly diminished and can in some species be reversed. The effect is more pronounced in the apical parts of the roots (11).

Mobilization of Iron in the Rhizosphere

The response of roots to iron deficiency is more elaborate, and for soil grown plants has been worked out, notably by Romheld and Marschner (12) in Germany. Most species growing in calcareous soils, in which the concentration of soluble iron is very low, respond by releasing hydrogen ions near the root apices, and also by enhancing iron reduction at the root surfaces. An example in Table 3.4 shows that the tendency of the roots of peanut plants to reduce iron is

TABLE 3.4. Rate of Fe^{III} Reduction by Roots of Peanut Plants Grown for 18 Days in Various Soils as Affected by Addition of FeEDDHA (10 mg Fe/kg soil) [after (12)].

Soil	$CaCO_3$ content (%)	pH ($CaCl_2$)	Rate of Fe^{III} reduction (nmol/g fwt/h) -FeEDDHA	+FeEDDHA
Sandy loam	-	3.15	10	10
Sandy loam (subsoil)	-	4.52	75	10
Loam	37.5	7.56	120	17
Loess (subsoil)	27.5	7.60	143	12
Amorphous $CaCO_3$	97.0	7.65	70	10

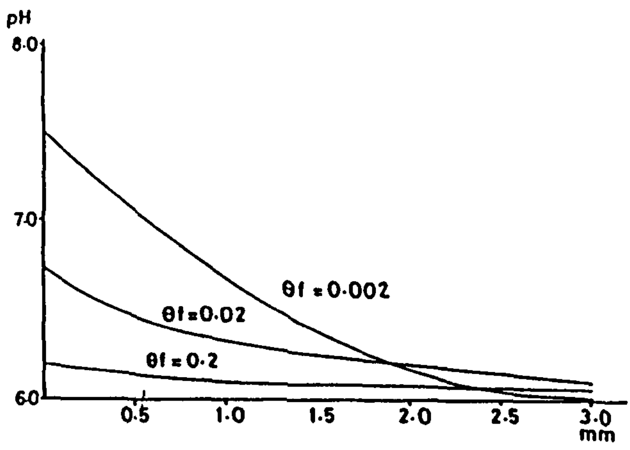

Figure 3.3 Effect of soil water content on the pH profile (HCO_3- release; time, 10^6s) (after Nye, 1981). HCO_3- diffuses much more readily through very moist soil (θf = 0.2) than moist soil (θf = 0.02), or dry soil (θf = 0.002).

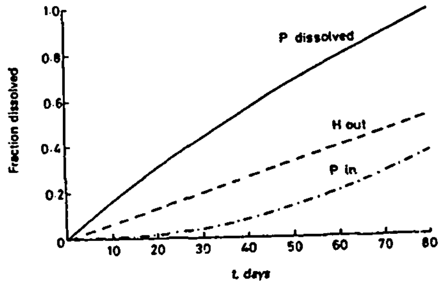

Figure 3.4 The effect of acid excretion by roots on the rate of dissolution and the uptake of P from fully substituted carbonate apatite. The rate of acid excretion is 3 x 10^{-11} mol dm^{-2} root surface s^{-1}. L_v = 1000 dm m^{-3}, a_r = 0.005 dm. This rate is typical of a well nodulated legume root system. If would add about 10 mmol dm^{-3} of acid to the soil in 100 days, giving, if no apatite dissolved, a reduction in the average soil pH of about 0.14 units.

____ fraction of P applied that has dissolved
---- fraction of OH- applied that has dissolved
-.-.- fraction of P applied that has been taken up

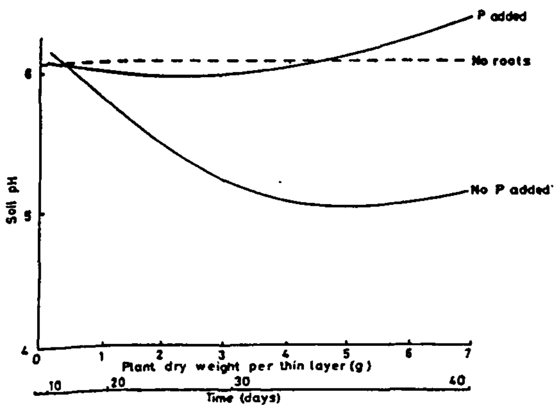

Figure 3.5 Effect of P on rape root rhizosphere pH (after Grinsted et al., 1982).

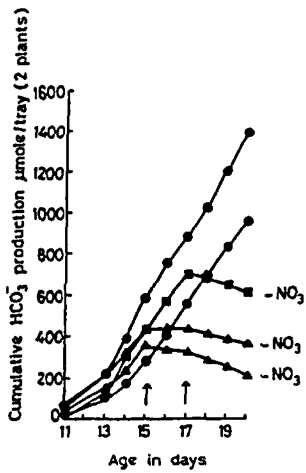

Figure 3.6 Effect of withdrawing NO_3- from nutrient culture solution on HCO_3- production from rape roots (after Moorby et al., 1985). Nitrate was withdrawn at day 17 (top - NO_3 curve) and day 15 (2 lower - NO_3 curves).

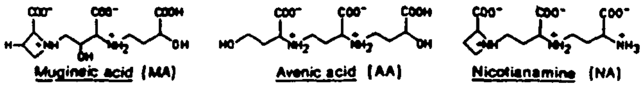

Figure 3.7 Structure of phytosiderophores and nicotianamine.

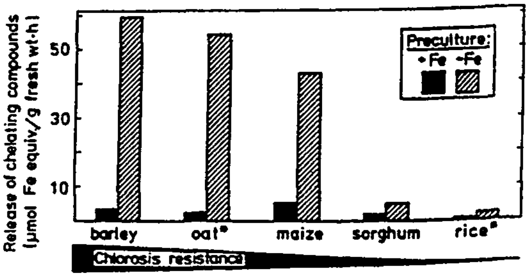

Figure 3.8 Effect of Fe nutritional status [preculture ± $Fe(OH)_3$] on the release of Fe-chelating compounds (phytosiderophores) by roots of grass species differing in resistance to Fe chlorosis (Kissel, unpublished; cited by Römheld and Marschner, 1986).

stimulated by alkaline conditions in the soil and low levels of soluble iron. They also exude more phenolic compounds that can form soluble complexes with iron and assist its transport through the soil to the surfaces. In contrast to this mode of action, grasses, but not monocots in general, are able to adopt a different strategy. The grasses appear to lack the mechanisms possessed by other species. Instead, they release so-called "phytosiderophores". These are amino acids that do not form the constituents of protein (Fig. 3.7). They resemble nicotinic acid, and have the ability to form stable soluble complexes with iron. Among the grasses resistance to iron chlorosis is correlated with the extent to which these compounds are released (Fig. 3.8). Here then, is an example of a very specific form of beneficial exudate.

Action of Ericoid Mycorrhizas

I come now to a specific microbiological mode of enhancing nutrient uptake. It is well known that the phosphate concentration of plants growing in phosphate-poor soils can be greatly increased by infection with endotrophic vesiculararbuscular mycorrhizas, such as *Endogone* sp. (13). They act by permeating the soil with hyphae that absorb phosphate at a distance from the root and convey it metabolically to the root itself, thus effectively increasing the uptake zone of the root. The ericoid mycorrhizas, which infect ericaceous species such as Rhodendendron, blueberry (*Vaccinium macrocarpon*) and heather (*Calluna vulgaris*), probably act in the same way; but they have an additional and much more important effect. The ericaceous species usually grow in very acid peaty soils, where mineral nitrogen is almost entirely in the ammonium form. I have for some time been puzzled to understand why such plants should apparently be making conditions even worse for themselves by exporting hydrogen ion in response to ammonium uptake. The work of Read (14,15) seems to me to go a long way towards answering this question. He has found that plants infected by ericoid mycorrhizas have an improved nitrogen status (Fig. 3.9) Absorption of ammonium by the roots lowers the pH to provide conditions favourable to the proteolytic action of enzymes associated with the mycorrhizas (Fig. 3.10) . Since the rate of nitrogen mineralisation is very low in these soils this action enhances the nitrogen uptake of the plants.

Competitive Ion Uptake

The effects mentioned so far concern nutrients in short supply in the rhizosphere. We have also to consider the interaction between nutrients, even when they are individually well supplied, because they may compete for uptake by the root. A good example is provided by calcium and magnesium. The concentration of each of these individually in the soil solution is usually adequate to convey them rapidly to the root surfaces, so there is no tranport limitation to their supply. The ratio of their concentrations in the plant does, however, depend on their concentration ratio in the soil solution; and on root preference, which is a plant physiological characteristic. When this is known, it is possible to predict the concentration ratio in a plant from the concentration ratio in the soil (16). Figure 3.11 compares the Ca/(Ca+Mg) ratio in the shoots of wheat plants with the corresponding ratio in a nutrient culture solution. It shows that the plants tend to take up Ca or Mg preferentially when the relative concentration of either ion in solution is low. Figure 3.12 shows how this plant physiological knowledge has been used to predict the Ca/(Ca+Mg) ratio of these plants when growing in soil. Since the balance between calcium and magnesium in forage plants is an important part of their feeding quality it is useful to know that it can be predictably controlled on any soil.

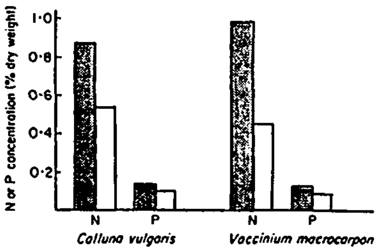

Figure 3.9 The nitrogen and phosphorus concentrations ericaceous species infected ■, and uninfected □, by ericoid mycorrhizas. (after Read, 1983).

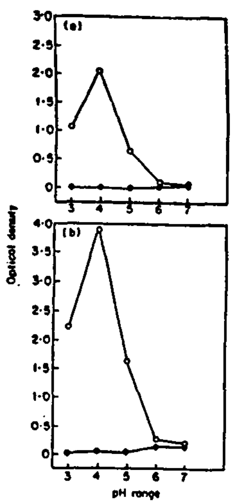

Figure 3.10 protease activity of mycorrhizal (O) and non-mycorrhizal (●) seedlings of *V. corymbosum* after (a) 3 d and (b) 6 d in buffered solutions containing a chromogenic substrate.

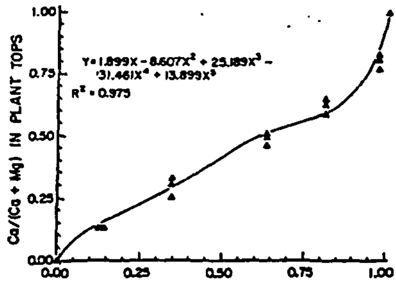

Figure 3.11 Ca/(Ca + Mg) ratios in nutrient solution-grown tops as a function of these ratios in nutrient solutions (after Rossi et al., 1988).

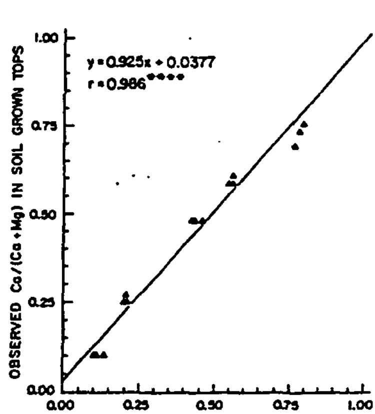

Figure 3.12 Correlation of observed Ca/(Ca + Mg) ratios in soil-grown tops with the ratios in tops predicted from Ca/(Ca + Mg) ratios in the soil solutions (after Rossi et al., 1988).

Root Shrinkage

Finally, I will touch on a physical effect. In the 1960s, there was much discussion about the movement of water from the soil to plant leaves, as a soil dried out. The main question was whether the resistance to flow when a plant began to wilt lay in the soil or in the conducting tissues of the plant. It was calculated that until the water potential in the soil dropped below about -1 bar, the hydraulic conductivity of the soil should be sufficiently high that there would be little difference in potential between the bulk of the soil and the root surfaces; consequently most of the resistance was thought to lie in the plant (17,18). In the early 1970s, Faiz and Weatherley (19,20,21) made an interesting experiment. They compared the water potential of the leaves of sunflower plants growing in solution culture and in soil. The soil was so moist that it should offer little resistance to the movement of water to the roots. They found that the leaf water potential was some 5 bar lower in the soil grown sunflower. When they vibrated the pots in which the plants were growing in soil on a shaker, the water potential in the leaves of the vibrated plants became about 5 bar greater than in unvibrated pots. This strongly suggested that there was a resistance to water movement at the soil-root interface itself, which was being alleviated by settling the soil around the root. If they contained the soil in plastic bags they obtained the same effect by squeezing the bags. We have recently measured the relation between the water potential in intact nodal roots of maize and shrinkage of the root diameter (Fig. 3.13). If we assume that finer roots show a similar relationship it is possible to estimate semi-quantitatively at what water potential an air gap is likely to develop between the soil and a root of given diameter over most of its circumference (Fig. 3.14) . Main roots (1-mm diam.) will lose full contact at a soil matric suction of about 0.2 bar. Fine roots (0.1-mm diam.) will lose contact at about 0.7 bar and root hairs (0.01-mm diam.) at about 2.3 bar. Thus over the course of a day's transpiration the main roots will lose contact first, then the fine roots and finally the root hairs. As the area of contact is progressively lost increased stress is imposed on the areas that remain. If the transport of water is restricted at the interface, so also will be the transfer of nutrients across it, and this is an effect we will have to take account of in future estimates of nutrient supply in drying soils.

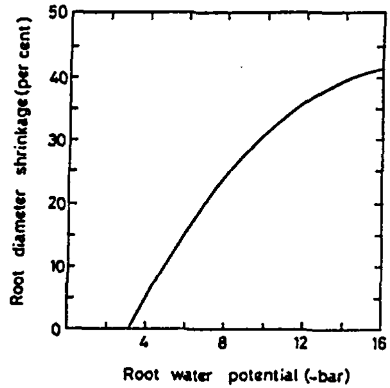

Figure 3.13 Relation between shrinkage (diameter) of major model roots and their water potential.

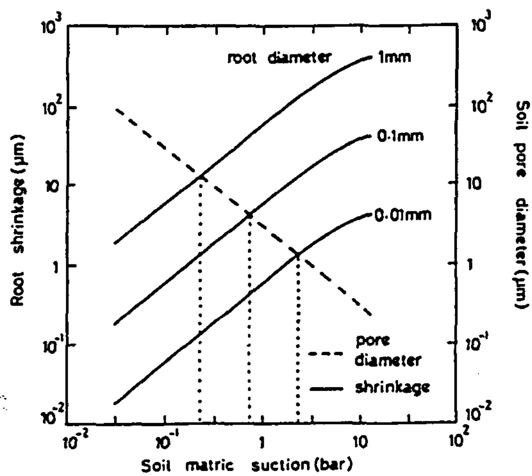

Figure 3.14 Initiation of an air-gap between root and soil in relation to root diameter and soil matric suction.

CONCLUSION

The diverse effects I have mentioned are a warning and a challenge. They warn us against adopting too simple a model of the way plants absorb nutrients from the soil. In many instances, particularly in reasonably fertile soils, good predictions of uptake of the major nutrients are now possible using simulation models, as I have illustrated in the first of these lectures. But for nutrients whose concentration in the soil solution is very low, plants can, by a variety of mechanisms, increase their uptake. These mechanisms become especially important when the plants are under stress as they are in infertile soils, and also when they are competing with one another in a plant community. Mechanistic understanding of the competition between plants for resources in the root zone, leading to prediction of the outcome, is virtually non-existent among soil scientists and ecologists. Therein lies a formidable challenge.

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